

Financial Econometrics

Binary dependent variable

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Overview

- ▶ Basics of binary dependent variables
- ▶ Linear probability model
- ▶ Logit and probit regression
- ▶ Comparing models and marginal effects
- ▶ Estimation and inference
- ▶ Measures of fit
- ▶ Hypothesis testing

Basics of binary dependent variables

Economic question

- ▶ Do banks discriminate on the basis of race when issuing loans?
- ▶ By law, they should not

Economic question

- ▶ Boston Fed HMDA Dataset ([link](#)): 2380 individual applications for single-family mortgages made in 1990 in the greater Boston area collected under the Home Mortgage Disclosure Act
- ▶ 28% of black applicants were denied mortgages, but only 9% of white applicants
- ▶ But, black and white applicants are not necessarily “identical but for their race”
- ▶ We need to compare rates of denials, *holding other applicant characteristics constant*
- ▶ Sounds like multiple regression, but with a twist
- ▶ **The dependent variable - loan denial - is binary**

Binary dependent variables

- ▶ Whether a loan application is accepted or not, is binary
- ▶ Many other important concerns have binary outcomes, such as
 - ▶ Whether a merger deal is paid in cash
 - ▶ Whether a household holds stocks
 - ▶ Whether a CEO is fired
 - ▶ ...

Discrimination in loan applications

- ▶ What other factors determine whether or not an individual loan application is denied?
- ▶ Size of the required loan payment relative to the applicant's income (P/I ratio)
- ▶ Load the data and investigate
- ▶ Scatterplot: x-axis = P/I ratio, y-axis = Denied

Packages you will need today

```
require(stargazer) # regression table  
require(lmtest) # robust SE  
require(sandwich) # robust SE  
require(DescTools) # winsorizing  
require(mfx) # marginal effects
```

Set-up the data

```
# read the data
load("Mortgage.RData")

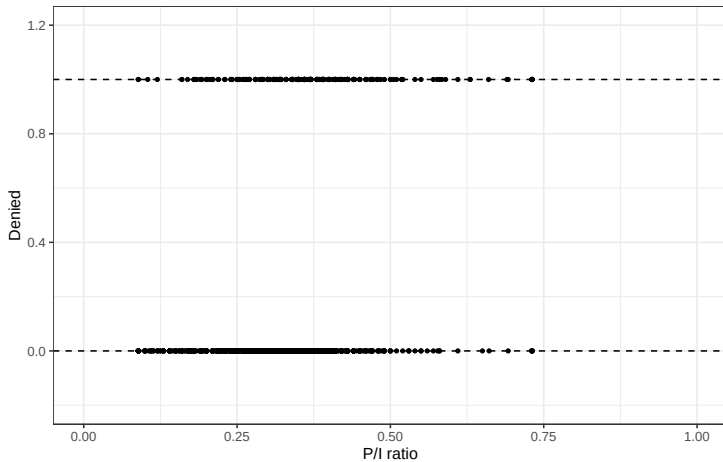
# clean observations with missing values
Mortgage <- na.omit(Mortgage)

# Re-scale and winsorize the main dependent variable
# Winsorizing leads to differences in the point estimates compared to the book
Mortgage$debt_income_ratio <- Winsorize(
  Mortgage$debt_income_ratio / 100,
  val=quantile(Mortgage$debt_income_ratio / 100,
    probs = c(0.005, 0.995), na.rm = T))

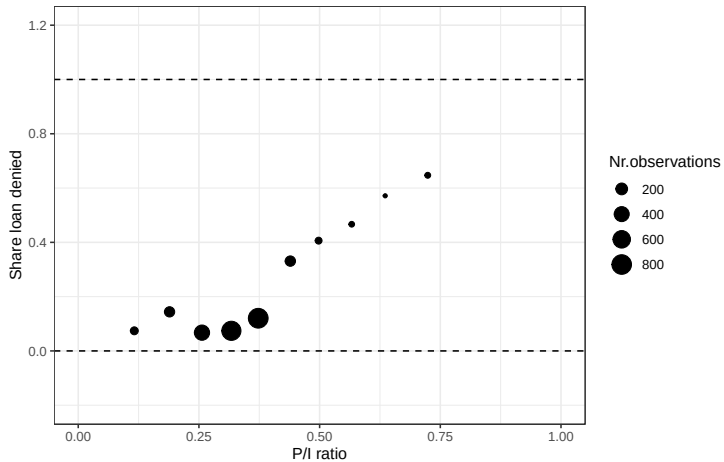
# inspect the data
str(Mortgage)

## 'data.frame':    2379 obs. of  8 variables:
## $ denied          : int  0 0 0 0 0 0 0 0 1 0 ...
## $ debt_income_ratio: num  0.221 0.265 0.372 0.32 0.36 ...
## $ black           : int  0 0 0 0 0 0 0 0 0 0 ...
## $ self_employed   : int  0 0 0 0 0 0 0 0 0 0 ...
## $ score_consumer   : int  5 2 1 1 1 1 1 2 2 2 ...
## $ score_mortgage   : int  2 2 2 2 1 1 2 2 2 1 ...
## $ insurance_denied : int  0 0 0 0 0 0 0 0 1 0 ...
## $ loan_price_ratio : num  0.8 0.922 0.944 0.881 0.6 ...
## - attr(*, "na.action")= 'omit' Named int 2246
## ..- attr(*, "names")= chr "2246"
```

Visualize the data



Visualize the data



Linear probability model

Linear probability model

Natural starting point is the linear regression model with a single regressor

$$D(\textit{denied}) = \alpha + \beta \textit{debt-to-income} + u$$

- What does the predicted value $\hat{y}$ mean when y is binary?

Lineary probability model

- ▶ In the LPM, the predicted value of y is interpreted as the predicted probability that $y = 1$, and β is the change in that predicted probability for a unit increase in x

$$y = \alpha + \beta x + u$$

- ▶ For binary y : $E(y|x) = 1 \times Prob(y = 1|x) + 0 \times Prob(y = 0|x) = Prob(y = 1|x)$
- ▶ Under zero conditional mean assumption ($E(u|x) = 0$):
 $E(y|x) = E(\alpha + \beta x + u) = \alpha + \beta x$
- ▶ So, $Prob(y = 1|x) = \alpha + \beta x$

Lineary probability model

- ▶ When y is binary, the linear regression model $y = \alpha + \beta x + u$ is called linear probabiltiy model because
$$Prob(y = 1|x) = \alpha + \beta x$$
- ▶ Predicted values from such models are probabilties and the relationship is linear
 - ▶ $\hat{y}$ is the predicted probability that $y = 1$, given x
 - ▶ β is the change in probability that $y = 1$ for a unit change in x

$$\beta_1 = \frac{Prob(y = 1|x + \Delta x) - Prob(y = 1|x)}{\Delta x}$$

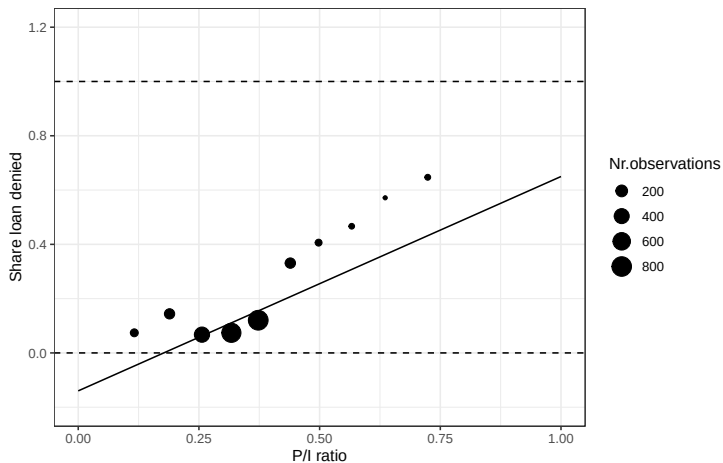
Lineary probability model applied to HMDA data

```
# regression
fit.lpm <- lm(denied ~ debt_income_ratio, data = Mortgage)

# robust standard errors
se.robust.fit.lpm <- coeftest(fit.lpm, vcov = vcovHC(fit.lpm))[,2]
# equivalent option: coeftest(fit.lpm, vcov = hccm)

# display the results
stargazer(list(fit.lpm),
           se = list(se.robust.fit.lpm),
           type="text", keep.stat=c("n", "rsq", "adj.rsq"),
           report=('vc*t'))
```


Linear probability model applied to HMDA data



Is there discrimination?

```
# regression
fit.lpm.multiple <- lm(denied ~ debt_income_ratio + black, data = Mortgage)

# robust standard errors
se.robust.fit.lpm.multiple <-
  coeftest(fit.lpm.multiple,
           vcov = vcovHC(fit.lpm.multiple))[,2]
# equivalent option: coeftest(fit.lpm.multiple, vcov = hccm)

# display the results
stargazer(list(fit.lpm, fit.lpm.multiple),
          se = list(se.robust.fit.lpm, se.robust.fit.lpm.multiple),
          type="text",
          keep.stat=c("n", "rsq", "adj.rsq"),
          report=('vc*t'))
```

Linear probability model applied to HMDA data

```
##
## =====
##                               Dependent variable:
##                               -----
##                               denied
##                               (1)           (2)
## -----
## debt_income_ratio    0.790***    0.712***
##                      t = 7.673    t = 6.944
##
## black                0.172***
##                      t = 6.900
##
## Constant             -0.140***    -0.139***
##                      t = -4.256    t = -4.261
##
## -----
## Observations         2,379        2,379
## R2                   0.041        0.075
## Adjusted R2          0.040        0.074
## =====
## Note:                *p<0.1; **p<0.05; ***p<0.01
```

Interpretation

$$\widehat{denied} = -0.139 + 0.712 \times debt_income_ratio + 0.172 \times black$$

- ▶ What is the predicted probability of loan denial for a black person with a $debt_income_ratio = 0.30$?
- ▶ $Prob(denied | debt_income_ratio = 0.30, black = 1) = -0.139 + 0.712 \times 0.3 + 0.172 \times 1 = 0.2466$
- ▶ What is the predicted probability of loan denial for a white person with the same $debt_income_ratio$?
- ▶ $Prob(denied | debt_income_ratio = 0.30, black = 0) = -0.139 + 0.712 \times 0.3 + 0.172 \times 0 = 0.0746$
- ▶ What's the effect of race? 17.2%
- ▶ Still plenty of room for omitted variable bias. . .

Summary linear probability model

- ▶ The LPM models $Prob(y = 1|x)$ as a linear function of x
- ▶ Advantages:
 - ▶ Simple to estimate and interpret
 - ▶ Inference the same as for multiple regression (note: LPM is inherently heteroskedastic)
- ▶ Disadvantages:
 - ▶ A LPM says that the change in the predicted probability for a given change in x is the same for all values of x , but that does not always make sense
 - ▶ Also, LPM predicts probabilities that can be < 0 and > 1
- ▶ Overall, we need a non-linear model: probit and logit regression

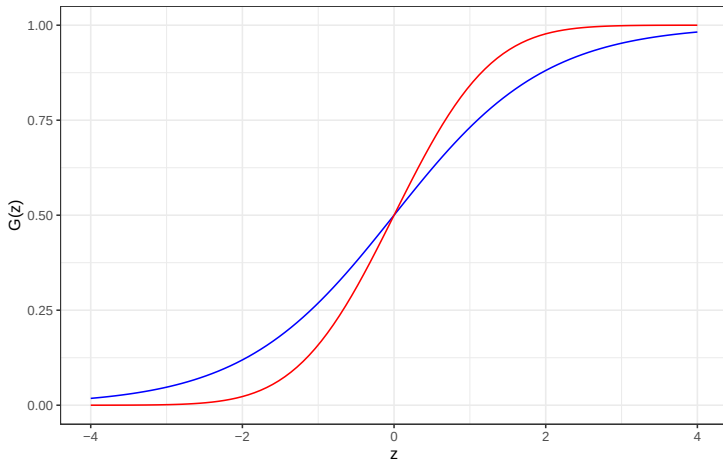
Logit and probit regression

Logit and probit regression

$$Prob(y = 1|x) = G(\alpha + \beta_1 x_1 + \cdots + \beta_k x_k)$$

- ▶ Probit model: $G(z) = \Phi(z) = \int_{-\infty}^z \phi(v) dv$ ($\Phi()$ is the standard normal cumulative distribution function)
- ▶ Logit model: $G(z) = \Lambda(z) = \frac{\exp(z)}{1 + \exp(z)}$ ($\Lambda(z)$ is the logistic cumulative distribution function)

Logit (blue) and probit (red) linking function, visualization



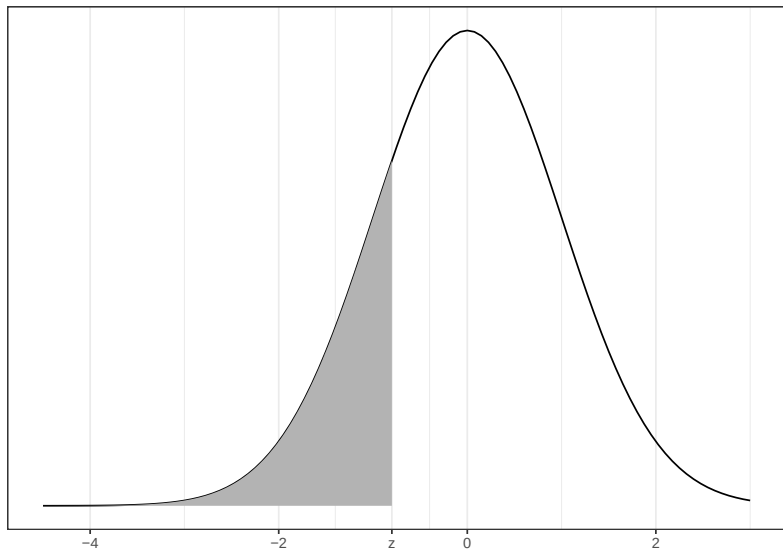
Probit regression

$$Prob(y = 1|x) = \Phi(\alpha + \beta_1 x_1 + \cdots + \beta_k x_k)$$

- ▶ Probit regression models the probability that $y = 1$ using the cumulative standard normal distribution, $\Phi(z)$ evaluated at $z = \alpha + \beta_1 x_1 + \cdots + \beta_k x_k$
- ▶ $\Phi(z) = \int_{-\infty}^z \phi(v) dv$
- ▶ $z = \beta_0 + \beta_1 x_1 + \cdots + \beta_k x_k$ is the “z-value” or “z-index”
- ▶ Suppose
 - ▶ $Prob(y = 1|x) = \Phi(\beta_0 + \beta_1 x)$
 - ▶ $\beta_0 = -2$, $\beta_1 = 3$, and $x = 0.4$
 - ▶ $Prob(y = 1|0.4) = \Phi(-2 + 3 \times 0.4) = \Phi(-0.8)$
 - ▶ $Prob(y = 1|0.4)$ = area under the standard normal density to the left of $z = -0.8$, which is 0.2119
 - ▶ in R, $\Phi(z)$ equals:

```
pnorm(z, mean = 0, sd = 1)
```

Probit regression



Why cumulative normal regression?

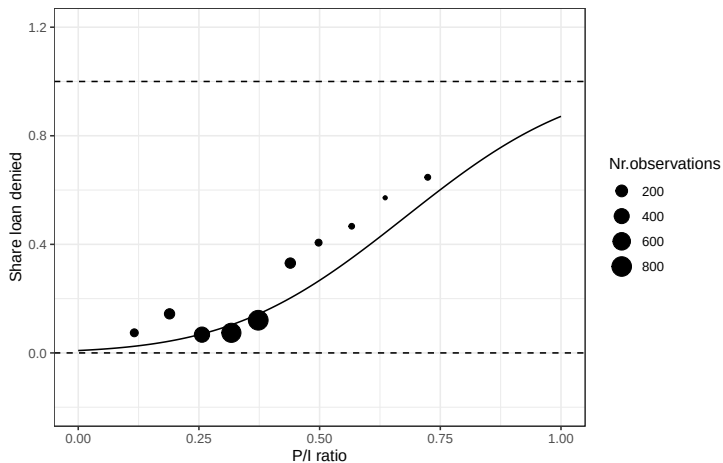
- ▶ The “S-shape” gives us:
 - ▶ Marginal probabilities depend on the value of x
 - ▶ Predicted probabilities are bounded between 0 and 1
- ▶ Easy to use
- ▶ Relative straight forward interpretation
 - ▶ $z = \beta_0 + \beta_1 x_1 + \dots + \beta_k x_k$
 - ▶ β_1 is the change in the z-value of a unit change in x_1 holding everything else constant

Probit applied to HMDA data

```
# regression
fit.probit <- glm(denied ~ debt_income_ratio, data = Mortgage,
                  family = binomial(link = probit))

# display the results
stargazer(list(fit.probit),
           se = list(se.robust.fit.lpm),
           type="text", keep.stat=c("n", "rsq", "adj.rsq"),
           report=('vc*t'))
```


Probit applied to HMDA data



Is there discrimination?

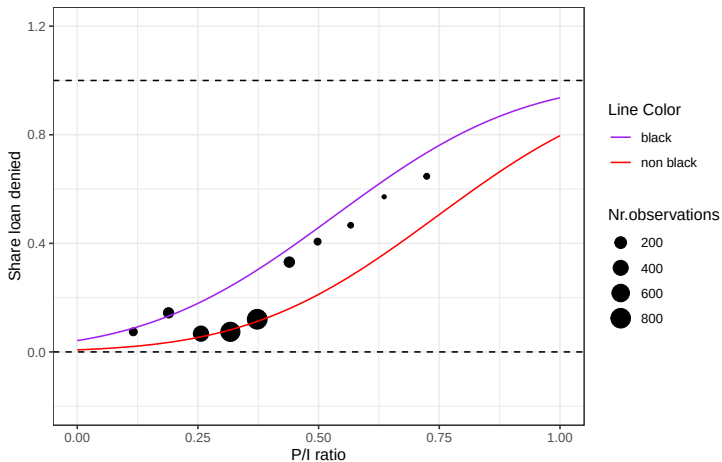
```
# regression  
fit.probit.multiple <- glm(denied ~ debt_income_ratio + black,  
                           data = Mortgage,  
                           family = binomial(link = probit))
```

```
# display the results  
stargazer(list(fit.probit, fit.probit.multiple),  
           se = list(se.robust.fit.lpm),  
           type="text",  
           keep.stat=c("n", "rsq", "adj.rsq"),  
           report=('vc*t'))
```

Probit applied to HMDA data

```
##
## =====
##                               Dependent variable:
##                               -----
##                               denied
##                               (1)           (2)
## -----
## debt_income_ratio    3.510***          3.258***
##                      t = 8.597          t = 7.856
##
## black                0.695***
##                      t = 8.318
##
## Constant             -2.376***          -2.429***
##                      t = -16.232        t = -16.304
##
## -----
## Observations         2,379              2,379
## =====
## Note:                *p<0.1; **p<0.05; ***p<0.01
```

Probit applied to HMDA data



Interpretation

$$\widehat{denied} = \Phi(-2.429 + 3.258 \times debt_to_income + 0.695 \times black)$$

- ▶ What is the predicted probability of loan denial for a black person with a $debt_income_ratio = 0.30$?
- ▶ $Prob(denied | \times debt_income_ratio = 0.30, black = 1) = \Phi(-2.429 + 3.258 \times 0.30 + 0.695 \times 1) = 0.225$
- ▶ What is the predicted probability of loan denial for a white person with the same $debt_income_ratio$?
- ▶ $Prob(denied | \times debt_income_ratio = 0.30, black = 0) = \Phi(-2.429 + 3.258 \times 0.30 + 0.695 \times 0) = 0.0733$
- ▶ What's the effect of race? 15.1%
- ▶ Is it the same for a $debt_income_ratio = 0.10$?
- ▶ Still plenty of room for omitted variable bias...

Logit regression

$$Prob(y = 1|x) = \Lambda(\alpha + \beta_1 x_1 + \cdots + \beta_k x_k)$$

- ▶ Logit regression models the probability that $y = 1$ using the cumulative standard logistic distribution function, $\Lambda(z)$ evaluated at $z = \beta_0 + \beta_1 x_1 + \cdots + \beta_k x_k$
- ▶ $\Lambda(z) = \frac{\exp(z)}{1 + \exp(z)}$
- ▶ Suppose
 - ▶ $Prob(y = 1|x) = \Lambda(\beta_0 + \beta_1 x)$
 - ▶ $\beta_0 = -2$, $\beta_1 = 3$, and $x = 0.4$
 - ▶ $Prob(y = 1|0.4) = \Lambda(-2 + 3 \times 0.4) = \Lambda(-0.8)$
 - ▶ $Prob(y = 1|0.4) = 0.31$
- ▶ Note: because logit and probit use different probability functions, the coefficients (β 's) are different in logit and probit
- ▶ In R, $\Lambda(z)$ equals:

```
plogis(z, location = 0, scale = 1)
```

Logit regression

$$Prob(y = 1|x) = \Lambda(\alpha + \beta_1 x_1 + \cdots + \beta_k x_k)$$

- ▶ Odds: $\frac{Prob(y=1|x)}{Prob(y=0|x)} = \frac{Prob(y=1|x)}{1-Prob(y=1|x)}$
- ▶ Odds in a logit model: $\exp(\alpha + \beta_1 x_1 + \cdots + \beta_k x_k)$
- ▶ The coefficients are given in units of log odds
- ▶ β_1 is the change in the log-odds for a unit change in x_1 holding everything else constant

Logit applied to HMDA data

```
# regression  
fit.logit <- glm(denied ~ debt_income_ratio,  
                 data = Mortgage,  
                 family = binomial(link = logit))
```

```
# display all models  
stargazer(list( fit.logit),  
           type="text",  
           keep.stat=c("n", "rsq", "adj.rsq"),  
           report=('vc*t'))
```


Is there discrimination?

```
# regression  
fit.logit.multiple <- glm(denied ~ debt_income_ratio + black,  
                           data = Mortgage,  
                           family = binomial(link = logit))
```

```
# display all models  
stargazer(list(fit.logit, fit.logit.multiple),  
           type="text",  
           keep.stat=c("n", "rsq", "adj.rsq"),  
           report=('vc*t'))
```

Logit applied to HMDA data

```
##
## =====
##                               Dependent variable:
##                               -----
##                               denied
##                               (1)           (2)
## -----
## debt_income_ratio    6.892***    6.452***
##                      t = 9.229    t = 8.539
##
## black                1.248***
##                      t = 8.508
##
## Constant             -4.377***    -4.492***
##                      t = -15.849   t = -15.963
##
## -----
## Observations         2,379        2,379
## =====
## Note:                *p<0.1; **p<0.05; ***p<0.01
```

Interpretation

$$\widehat{denied} = \Lambda(-4.492 + 6.452 \times debt_to_income + 1.248 \times black)$$

- ▶ What is the predicted probability of loan denial for a black person with a $debt_income_ratio = 0.30$?
- ▶ $Prob(denied | \times debt_income_ratio = 0.30, black = 1) = \Lambda(-4.492 + 6.452 \times 0.30 + 1.248 \times 1) = 0.213$
- ▶ What is the predicted probability of loan denial for a white person with the same $debt_income_ratio$?
- ▶ $Prob(denied | \times debt_income_ratio = 0.30, black = 0) = \Lambda(-4.492 + 6.452 \times 0.30 + 1.248 \times 0) = 0.072$
- ▶ What's the effect of race? 14.1%
- ▶ Is it the same for a $debt_income_ratio = 0.10$?
- ▶ Still plenty of room for omitted variable bias...

Logit versus Probit

- ▶ The main reason is historical: logit is computationally faster and easier, but that does not matter nowadays
- ▶ In practise, logit and probit are very similar
- ▶ Empirical results typically should not hinge on the logit versus probit choice

Quick Recap

- ▶ Basics of Binary Dependent Variables
- ▶ Linear Probability Model
- ▶ Logit and Probit Regression
- ▶ Comparing Models & Marginal Effects
- ▶ Estimation and Inference
- ▶ Measures of Fit
- ▶ Hypothesis Testing

Comparing Models & Marginal Effects

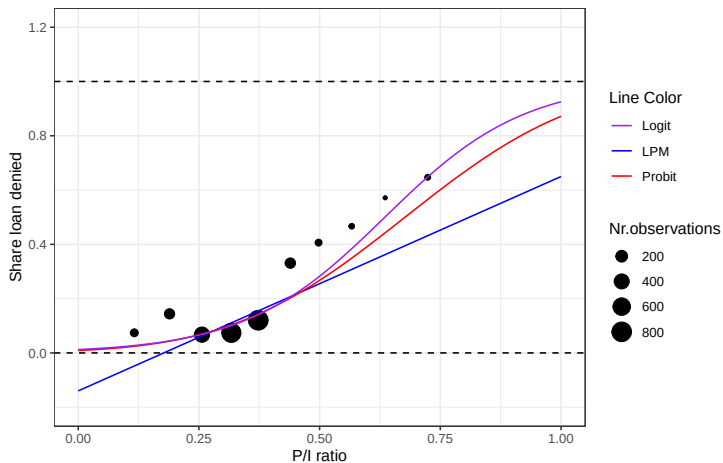
Comparing all three models

```
stargazer(list(fit.lpm.multiple,
               fit.logit.multiple,
               fit.probit.multiple),
           se = list(se.robust.fit.lpm.multiple,
                     NULL, NULL),
           type="text",
           keep.stat=c("n", "rsq", "adj.rsq"),
           report=('vc*t'))
```

Comparing all three models

```
##
## =====
##                               Dependent variable:
##                               -----
##                               denied
##                               OLS      logistic      probit
##                               (1)      (2)      (3)
## -----
## debt_income_ratio  0.712***    6.452***    3.258***
##                   t = 6.944    t = 8.539    t = 7.856
##
## black              0.172***    1.248***    0.695***
##                   t = 6.900    t = 8.508    t = 8.318
##
## Constant           -0.139***   -4.492***   -2.429***
##                   t = -4.261 t = -15.963 t = -16.304
##
## -----
## Observations       2,379        2,379        2,379
## R2                  0.075
## Adjusted R2        0.074
## =====
## Note:                *p<0.1; **p<0.05; ***p<0.01
```

Predicted path for all models



Marginal effects

- ▶ Marginal effect: the effect on the dependent variable that results from changing an independent variable by a small amount
- ▶ Probit and logit are non-linear functions
- ▶ Hence, the ultimate effect of one unit change in a regressor (x) on predicted probabilities is different from different values of x
- ▶ It is common to report marginal (or partial) effects instead of coefficients
 - ▶ Marginal effect at the means
 - ▶ Average marginal effect

Marginal effects

```
# marginal effect at the mean
fit.probit.mfx.at.mean <-
  probitmfx(denied ~ debt_income_ratio + black,
            data = Mortgage,
            atmean = T)

# average marginal effect
fit.probit.mfx.average <-
  probitmfx(denied ~ debt_income_ratio + black,
            data = Mortgage,
            atmean = F)
```

Marginal effects

```
fit.probit.mfx.at.mean
```

```
## Call:
## probitmfx(formula = denied ~ debt_income_ratio + black, data
##          atmean = T)
##
## Marginal Effects:
##              dF/dx Std. Err.      z    P>|z|
## debt_income_ratio 0.588885  0.073852 7.9738 1.538e-15 ***
## black              0.166484  0.024505 6.7939 1.091e-11 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## dF/dx is for discrete change for the following variables:
##
## [1] "black"
```

Marginal effects

```
fit.probit.mfx.average
```

```
## Call:
## probitmfx(formula = denied ~ debt_income_ratio + black, data
##      atmean = F)
##
## Marginal Effects:
##              dF/dx Std. Err.      z    P>|z|
## debt_income_ratio 0.597761  0.075365  7.9316 2.164e-15 ***
## black              0.165578  0.024077  6.8769 6.116e-12 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## dF/dx is for discrete change for the following variables:
##
## [1] "black"
```

Estimation and Inference

Estimation of Logit and Probit

- ▶ Focus on probit with one explanatory variable
- ▶ Minimize squared error
 - ▶ Recall OLS: $\min \sum_{i=1}^n (y_i - \alpha - \beta x)^2 \rightarrow \alpha, \beta$
 - ▶ Non-linear least squares $\min \sum_{i=1}^n (y_i - \Phi(\alpha - \beta x))^2 \rightarrow \alpha, \beta$
 - ▶ How to solve minimization problem?
 - ▶ specialized minimization algorithm
 - ▶ Instead of non-linear least squares, a more efficient estimator is used. . .

Maximum Likelihood

- ▶ The likelihood function is the conditional density of $y_1, \dots, y_n$ given $x_1, \dots, x_n$ treated as a function of the unknown parameters α, β
- ▶ Maximizing the probability of observing the data given the assumed model
- ▶ For probit (with one explanatory variable):

$$f(y_i|x_i, \alpha, \beta) =$$

$$\Phi(\alpha + \beta x_i)^{y_i} \times (1 - \Phi(\alpha + \beta x_i))^{1-y_i}$$

$$\ln(L(\alpha, \beta)) = \ln\left(\prod_{i=1}^n f(y_i|x_i, \alpha, \beta)\right)$$

$$\max \ln(L(\alpha, \beta)) \rightarrow \alpha, \beta$$

Maximum likelihood

- ▶ The maximum likelihood estimator (MLE) is the value of $\alpha, \beta_1, \dots, \beta_k$ that maximize the likelihood function
- ▶ The MLE is the value of $\alpha, \beta_1, \dots, \beta_k$ that best describe the full distribution of the data
- ▶ In large samples, MLE is:
 - ▶ consistent
 - ▶ normally distributed
 - ▶ efficient (has the smallest variance of all estimators)

Measures of fit

Measures of fit for logit and probit

- ▶ The R^2 and *Adjusted R^2* do not make sense here (even for LPM). So, two other specialised measures are used
 - ▶ The fraction correctly predicted = fraction of y 's for which the predicted probability is $> 50\%$ when y_i is 1, or is $< 50\%$ when y_i is 0
 - ▶ What about rare events (rare disease)? Ease to predict $y = 0$, but hard to predict $y = 1$. How does the aforementioned measure behave?
 - ▶ The pseudo R^2 (also known as McFadden R^2) measures the improvement in the value of the log likelihood, relative to having no x 's: $1 - \frac{\ln(L(\alpha, \beta_1, \dots, \beta_k))}{\ln(L(\alpha))}$

Measures of fit

```
# Probit
```

```
PseudoR2(fit.probit)
```

```
##      McFadden
```

```
## 0.04813484
```

```
# Equivalent approach
```

```
1 - fit.probit$deviance / fit.probit$null.deviance
```

```
## [1] 0.04813484
```

```
# Logit
```

```
PseudoR2(fit.logit)
```

```
##      McFadden
```

```
## 0.05159848
```

Hypothesis testing

Hypothesis test

- ▶ Usual t -tests and confidence intervals can be used testing $H_0: \beta=0$
- ▶ For joint hypothesis test, use likelihood ratio test that compares likelihood functions of the restricted and unrestricted models
- ▶ For q restrictions:

$$LR = 2(\ln(L(\text{unrestricted})) - \ln(L(\text{restricted}))) \sim \chi_q^2$$

- ▶ Our example:
 - ▶ we augment previous model with following variables:
score_consumer and score_mortgage
 - ▶ subsequently we ask, whether coefficients on both variables are jointly zero

Hypothesis test

```
# unrestricted
fit.probit.unrestricted <- glm(
  denied ~ debt_income_ratio + black + score_consumer + score_mortgage,
  data = Mortgage, family = binomial(link = probit))
# restricted
fit.probit.restricted <- glm(denied ~ debt_income_ratio + black,
                             data = Mortgage,
                             family = binomial(link = probit))
# likelihood ratio test (require lmtest)
lrtest(fit.probit.restricted, fit.probit.unrestricted)
```

```
## Likelihood ratio test
```

```
##
```

```
## Model 1: denied ~ debt_income_ratio + black
```

```
## Model 2: denied ~ debt_income_ratio + black + score_consumer + score_mortgage
```

```
##   #Df  LogLik Df  Chisq Pr(>Chisq)
```

```
## 1    3 -796.75
```

```
## 2    5 -746.07  2 101.35 < 2.2e-16 ***
```

```
## ---
```

```
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Summary

- ▶ If y is binary, then $E(y|x) = Prob(y = 1|x)$
- ▶ Three models
 - ▶ linear probability model (linear multiple regression)
 - ▶ probit (cumulative standard normal distribution)
 - ▶ logit (cumulative standard logistic distribution)
- ▶ All models produce predicted probabilities
- ▶ Probit and logit are estimated using maximum likelihood
- ▶ Marginal effects depend on initial values of x for logit and probit

Textbook

- ▶ Binary dependent variable: Chapter 11, Stock and Watson, Introduction to Econometrics, Global Edition, 4th edition